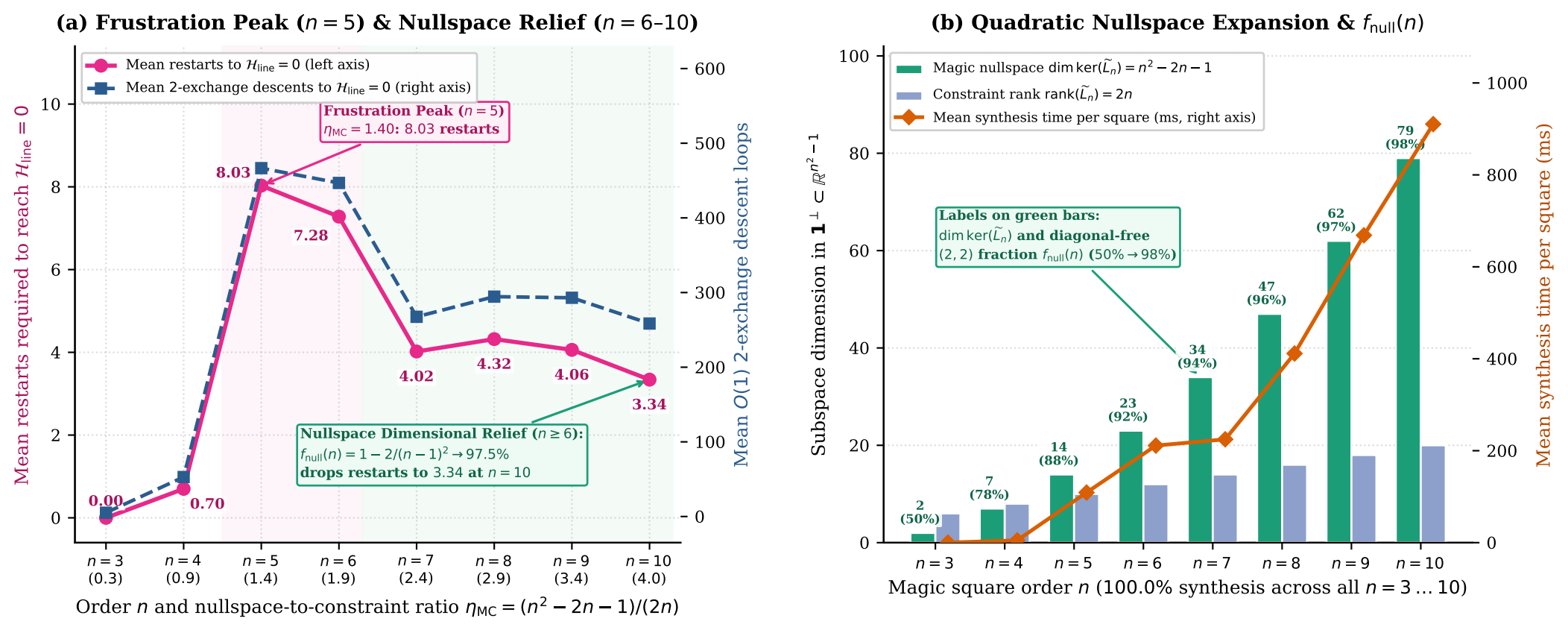

(a) Frustration Peak ($n = 5$) & Nullspace Relief ($n = 6$–10)
Mean restarts to $\mathcal{H}_{\text{line}} = 0$ (left axis)
Mean 2-exchange descents to $\mathcal{H}_{\text{line}} = 0$ (right axis)
Frustration Peak ($n = 5$)
$\eta_{\text{MC}} = 1.40$: 8.03 restarts
Nullspace Dimensional Relief ($n \geq 6$):
$f_{\text{null}}(n) = 1 - 2/(n-1)^2 \to 97.5\%$
drops restarts to 3.34 at $n = 10$
0.00
0.70
8.03
7.28
4.02
4.32
4.06
3.34
Mean restarts required to reach $\mathcal{H}_{\text{line}} = 0$
Mean $O(1)$ 2-exchange descent loops
$n = 3$ (0.3)
$n = 4$ (0.9)
$n = 5$ (1.4)
$n = 6$ (1.9)
$n = 7$ (2.4)
$n = 8$ (2.9)
$n = 9$ (3.4)
$n = 10$ (4.0)
Order n and nullspace-to-constraint ratio $\eta_{\text{MC}} = (n^2 - 2n - 1)/(2n)$
(b) Quadratic Nullspace Expansion & $f_{\text{null}}(n)$
Magic nullspace $\dim \ker(\widetilde{L}_n) = n^2 - 2n - 1$
Constraint rank $\text{rank}(\widetilde{L}_n) = 2n$
Mean synthesis time per square (ms, right axis)
Labels on green bars:
$\dim \ker(\widetilde{L}_n)$ and diagonal-free
$(2, 2)$ fraction $f_{\text{null}}(n)$ (50% → 98%)
2 (50%)
7 (78%)
14 (88%)
23 (92%)
34 (94%)
47 (96%)
62 (97%)
79 (98%)
Subspace dimension in $\mathbf{1}^{\perp} \subset \mathbb{R}^{n^2-1}$
Mean synthesis time per square (ms)
Magic square order n (100.0% synthesis across all $n = 3 \ldots 10$)